Problem 15.1

Consider an investment of \$5,000 at 6% convertible semiannually. How much can be withdrawn each half-year to use up the fund exactly at the end of 20 years?

Solution

The nominal interest rate is 6% convertible semiannually, so the effective interest rate per half-year is

$$i = \frac{0.06}{2} = 0.03.$$

Since withdrawals are made every half-year for 20 years, the number of withdrawals is

$$n = 20(2) = 40.$$

Let x be the amount withdrawn each half-year. The present value of the withdrawals must equal the initial investment of 5,000, so

$$5000 = x a_{\overline{40}|}.$$

Using the formula for an annuity-immediate,

$$a_{\overline{40}|} = \frac{1 - v^{40}}{i},$$

where

$$v = \frac{1}{1+i} = \frac{1}{1.03}.$$

Thus,

$$5000 = x \left(\frac{1 - (1.03)^{-40}}{0.03} \right).$$

Solving for x,

$$x = \frac{5000}{\frac{1 - (1.03)^{-40}}{0.03}}.$$

Therefore,

$$x \approx 216.311.$$

Approach B

We can also solve the problem using accumulated values at the end of the 40 half-year periods. The initial investment accumulates to

$$5000(1.03)^{40}.$$

The withdrawals form an annuity-immediate accumulated to the end of the period, so

$$5000(1.03)^{40} = x s_{\overline{40|}}.$$

Using

$$s_{\overline{40|}} = \frac{(1.03)^{40} - 1}{0.03},$$

we get

$$5000(1.03)^{40} = x \left(\frac{(1.03)^{40} - 1}{0.03} \right).$$

Solving for x,

$$x = \frac{5000(1.03)^{40}}{\frac{(1.03)^{40} - 1}{0.03}}.$$

Thus,

$$x \approx 216.311.$$

Both approaches give the same result. Hence, the amount that can be withdrawn each half-year is

$$\boxed{216.31}.$$